

Course Code: CSA1510

Course Name: cloud computing

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Assessment No: CO2_AT3

Q1. How do Auto Scaling and Load Balancing handle high traffic, reduce downtime, and improve user experience?

Explain with a real-world example.

Sol:

During Seasonal Sales such as Amazon Great Indian Festival or Flipkart Big Billion Days, the number of users increases suddenly. cloud computing uses Auto Scaling and Load Balancing to handle this high traffic efficiently.

* Auto Scaling

It automatically increases the number of servers when traffic is high and removes extra servers when traffic decreases.

* Load Balancing

* It distributes incoming user requests evenly among available servers so that no single server becomes overloaded.

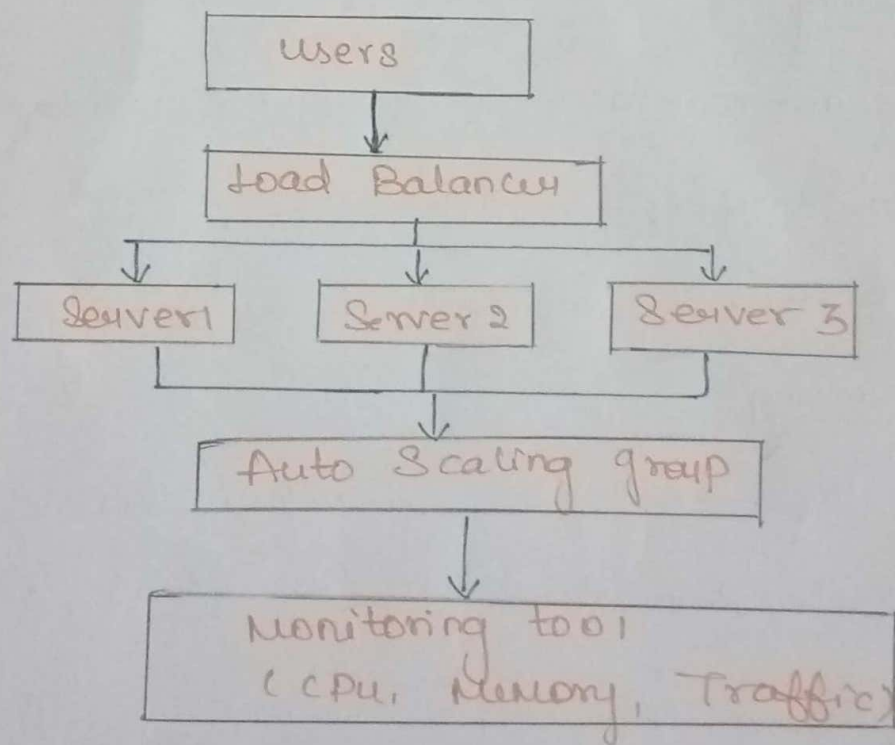
* This reduces downtime, prevents server crashes, and provides a smooth shopping experience.

- * Monitoring tools such as Amazon cloud watch continuously monitor cpu usage, memory, network traffic. when the load crosses a threshold, they trigger auto-scaling automatically

Real-world Example:

During the Amazon great Indian Festival, Auto Scaling adds extra servers during high traffic, while the load balancer distributes user requests evenly.. After the sale, the extra servers are removed to reduce costs.

Diagram



Benefits:

- * High availability
- * Reduced downtime
- * Better user experience
- * Cost optimization

Q2. How do cloud storage solutions ensure secure, durable, and cost-effective backup with a disaster recovery example?

Sol:

Organizations store backup data on cloud storage services such as Amazon S3, Azure Blob Storage, or Google cloud storage.

Durability:

It ensures data is not lost even if hardware fails.

Data Redundancy:

It stores multiple copies of data across different servers or locations.

Encryption:

Data at Rest - Encrypts stored data

Data in Transit - Encrypts data during transfer

Access Control:

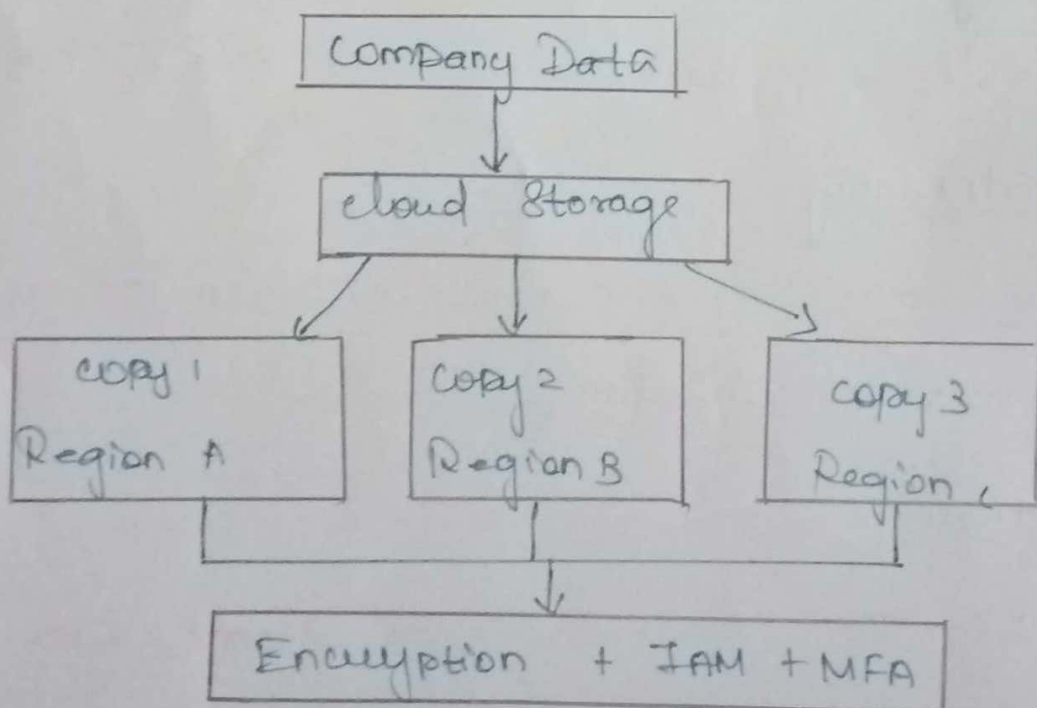
- * IAM
- * MFA
- * RBAC

cloud storage is more cost-effective because organizations pay only for storage they use without purchasing expensive hardware.

Disaster Recovery Example:

A company's local data center is damaged by a flood. Since backups are stored on Amazon S3 across multiple regions, the company restores its data quickly and resumes operations.

Diagram:



Benefits:

- * High durability
- * Secure backups
- * Lower storage cost
- * faster disaster recovery

Q3. How do CDNs and edge servers improve performance, reduce latency, provide global availability? Explain with an example.

Sol:

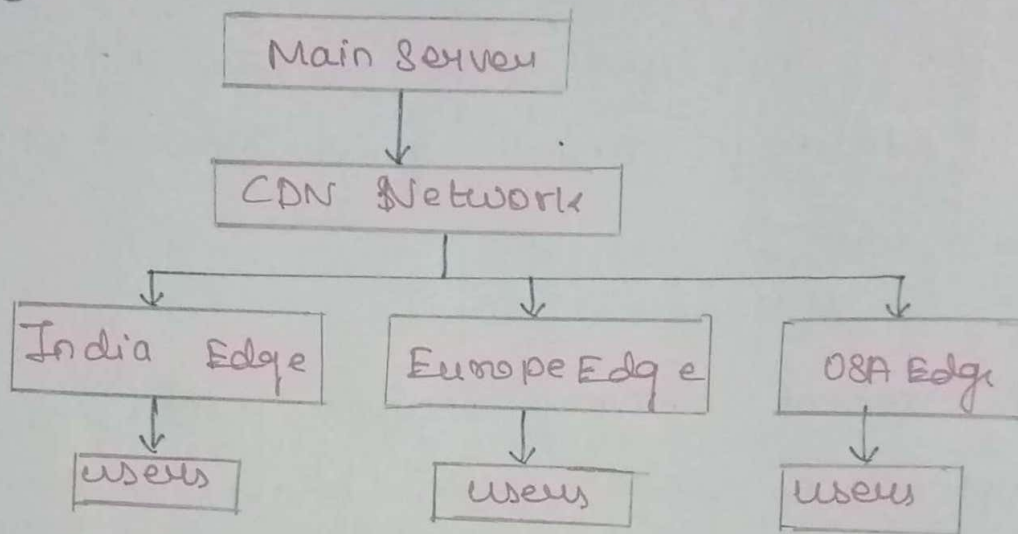
A Content Delivery Network (CDN) stores website content on edge servers located around the world.

- * users receive content from the nearest edge server instead of the main server.
- * This reduces latency, improves loading speed, and provides a better user experience.
- * cloud providers like AWS CloudFront, Azure CDN, Google Cloud CDN support global content delivery.

Real - world Example:

Netflix stores videos on edge servers worldwide. A user in India watches content from the nearest CDN server. Instead of the US data center, reducing buffering.

DIAGRAM:



Benefits:

- * Faster content delivery
- * Reduced latency
- * less Buffering
- * Global availability.

Q4. How do cloud providers ensure compliance, security, logging, monitoring, encryption, and data sovereignty in financial institutions?

801:

Banks require strict security and regulatory compliance when using cloud services.

cloud providers support compliance through standards such as:

* ISO 27001

* PCI-DSS

* GDPR

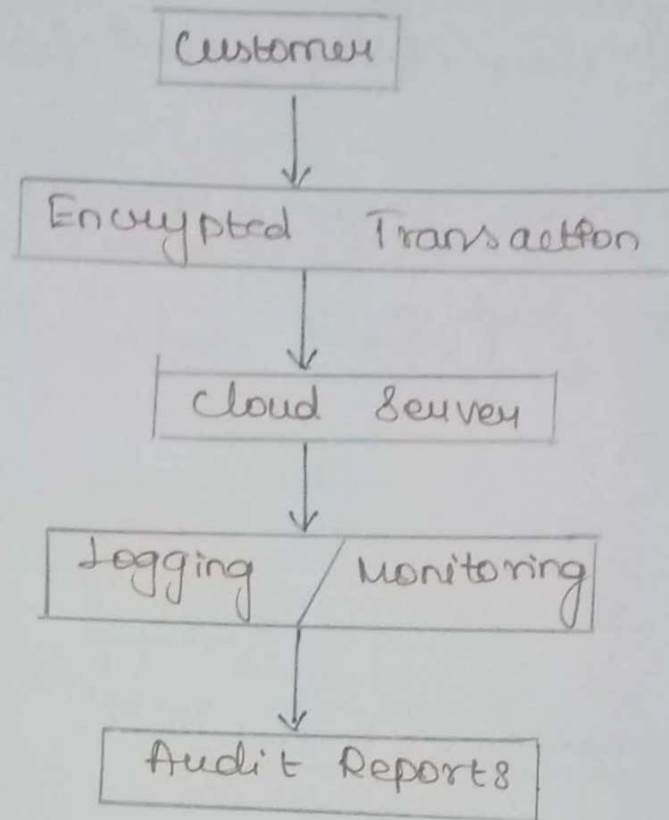
Security Mechanisms Include:

- * Logging - Records every user activity
- * Monitoring - Detects suspicious behavior
- * Encryption - Protects sensitive financial data.
- * Data Sovereignty - Ensures customer data remains within the required country or region.

Example:

A Bank encrypts online transaction data. Every transaction is logged for auditing. Customer data is stored in Indian cloud regions to comply with RBI regulations.

DIAGRAM:



Benefits:

- * Secure Transactions
- * Easy auditing
- * Regulatory compliance
- * customer trust.

Q5. How do containers differ from virtual machines, and how does Kubernetes simplify deployment, scaling, and portability? Explain with a microservices example.

801: Containers Package an applications along with its dependencies, making deployment faster and more efficient than virtual machines.

Virtual Machine	Container
Has full operating system	Shares host OS
More resource usage	less resource usage
Slow startup	Fast startup
larger size	lightweight.

Kubernetes is a container orchestration Platform that:

- * Automatically deploys containers
- * Scales applications when traffic increases
- * Restarts failed containers automatically
- * Balances workloads across servers.

Containers are portable and can run on AWS, Azure, google cloud, or on-premises without modification.

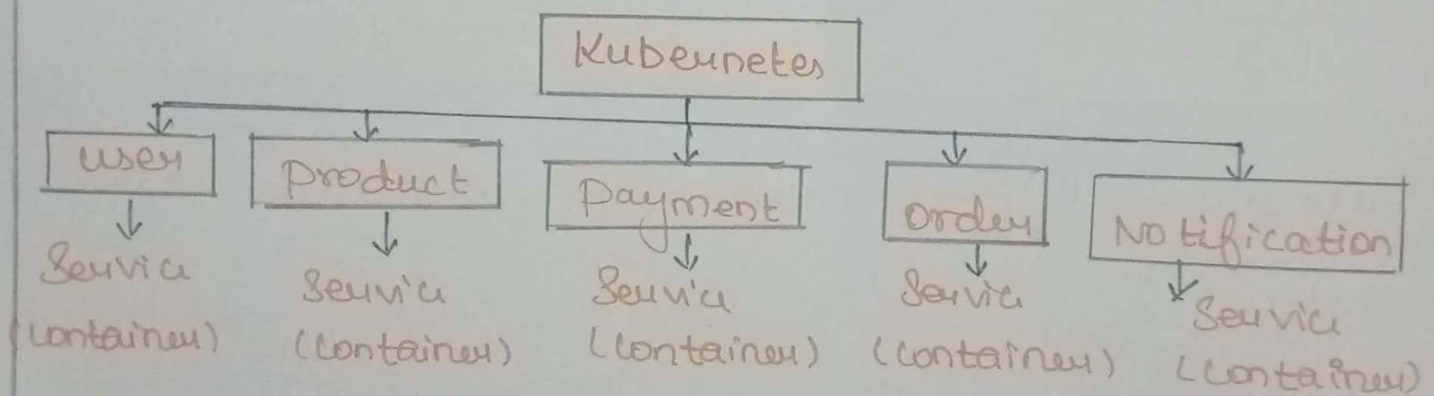
Microservices Example:

An online Shopping Applications:

- * User Service
- * Product Service
- * Payment Service
- * Order Service
- * Notification Service.

Each Service run in its own container.

Diagram:



Benefits:

- * Fast deployment
- * Better Scalability
- * High Portability
- * Efficient Resource utilization.